Dataset guide: summary

Sessions: 54. People (distinct IDs): 33, of whom 33 appear in more than one session. Recorded decisions: 21271.

Players per session: 4 to 8. Rounds: up to 3.

Choices modeled: do nothing, 1, 2, 3, 4. Reference choice (all weights are log−odds relative to it): do nothing.

Classes of player found: 7. class_1: 9 person(s); class_2: 6 person(s); class_3: 6 person(s); class_4: 3 person(s); class_5: 3 person(s); class_6: 3 person(s); class_7: 3 person(s)

Session settings examined as possible treatments: 34. Used as predictors: defendCost.

Patch−choice model: 29937 picks/stops from 9720 turns; 20 patch−choice features; pooled mean log−likelihood per pick −1.72 vs −3.39 for choosing uniformly. Treatments: treatment crossings used in patch choice: p_fine:‘protected?‘, defendCost:avail_1, defendCost:avail_4, defendCost:avail_2, fine:harvestYield, defendCost:‘cropped?‘, defendCost:avail_3, defendCost:cropOwner_mine.

CAUTIONS (detected automatically):

− How the classes were formed: 7 class(es) by recursive splitting (a class holding over 40% of people is split again when a split reaches mean silhouette 0.15; accepted splits scored 0.48, 0.37); people grouped by which patches they go for. Classes come from clustering noisy per−person weights, so treat them as a description of this sample.

How to read this guide

A WEIGHT is a coefficient from a multinomial logit: how much a feature raises (positive) or lowers (negative) the log−odds of a choice, relative to the reference choice. A weight of 0.7 roughly doubles the odds; −0.7 roughly halves them, other things equal.

A CLASS is a group of people whose individual models would choose alike in the same circumstances. By default (class_basis = 'choice') that means which patches they go for next, or whether they stop (the patch−choice model, Step 6b); the action they take there is described separately. A class describes how people respond to the board and game state in this data; it is not a personality type. WHEN a person acts is invisible unless round/phase were switched on in variableRoles.csv.

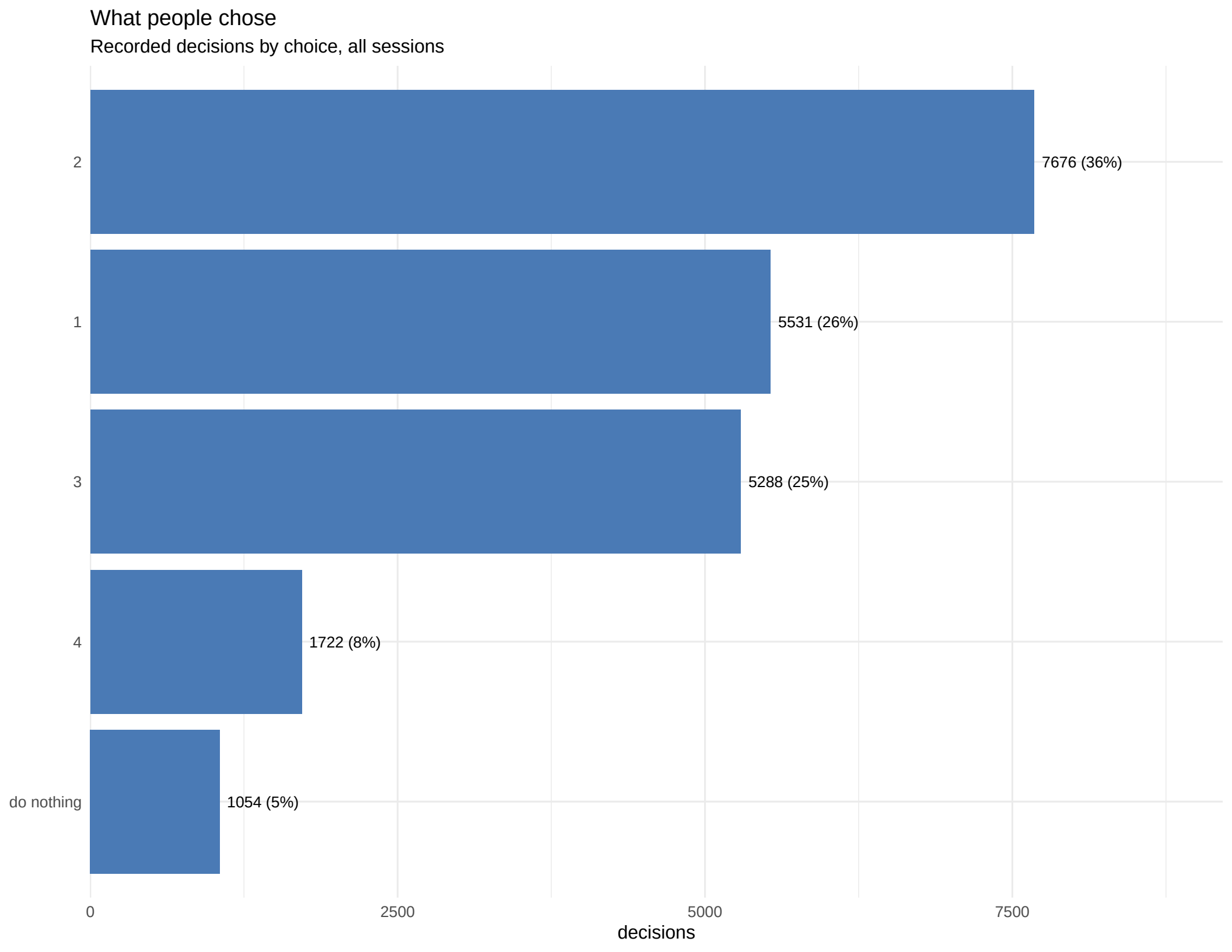
PATCH−CHOICE weights (the 'where' heat map) are per standard deviation of each feature: blue = patches with more of it are more likely to be picked next, red = less likely. Rows named stop and stop_k describe the tendency to end the turn, and how it grows with the number of picks already made. cropOwner_mine (or similar) means the patch holds the player's own crop; avail_N means action N is possible there. A name like p_fine:protected? is a TREATMENT crossing: how a session setting changes the appeal of that kind of patch (negative: patches with it become less likely to be picked as the setting rises).

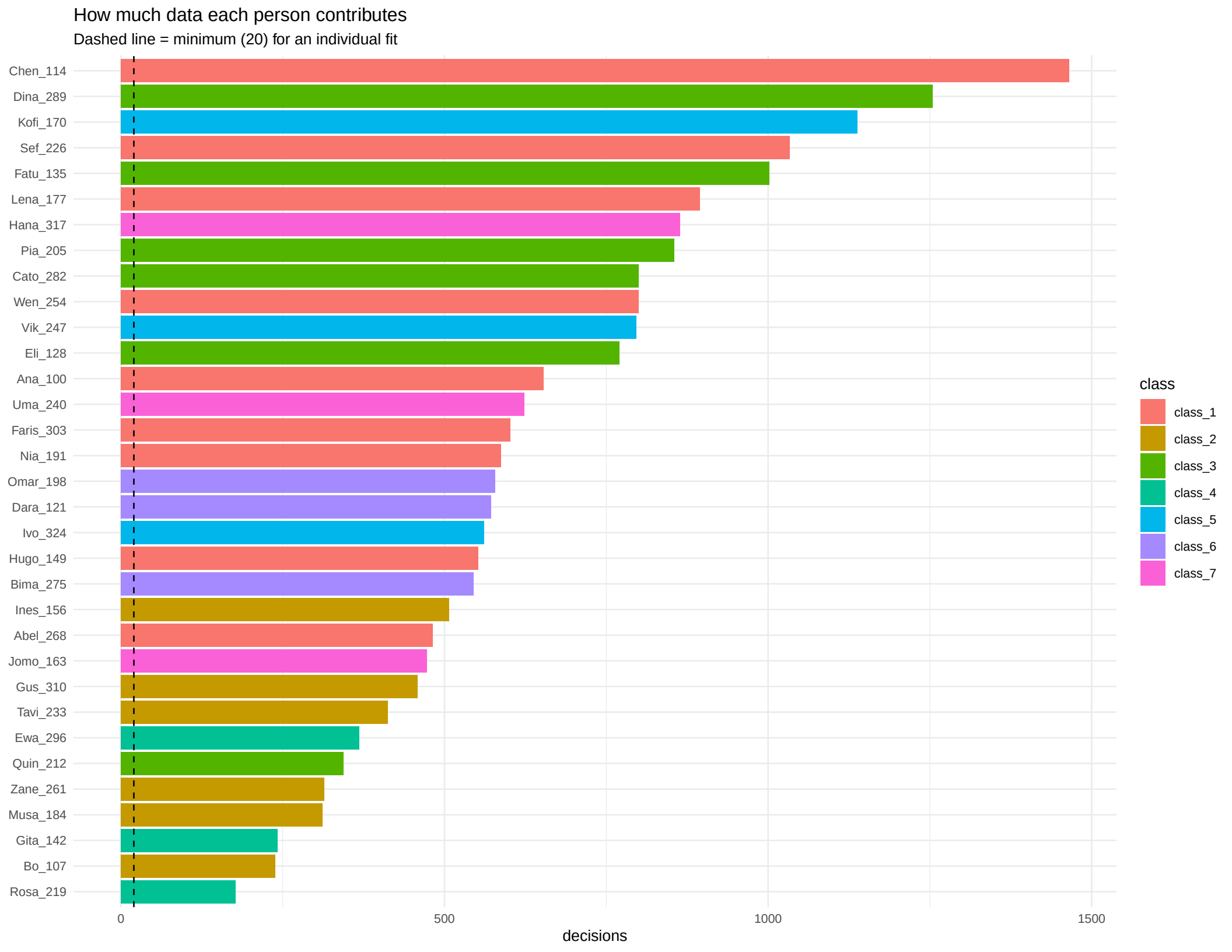
Heat maps: rows are features, columns are choices, colour shows the weight (clipped for readability; the printed number is the true value). Red lowers the chance of that choice, blue raises it.

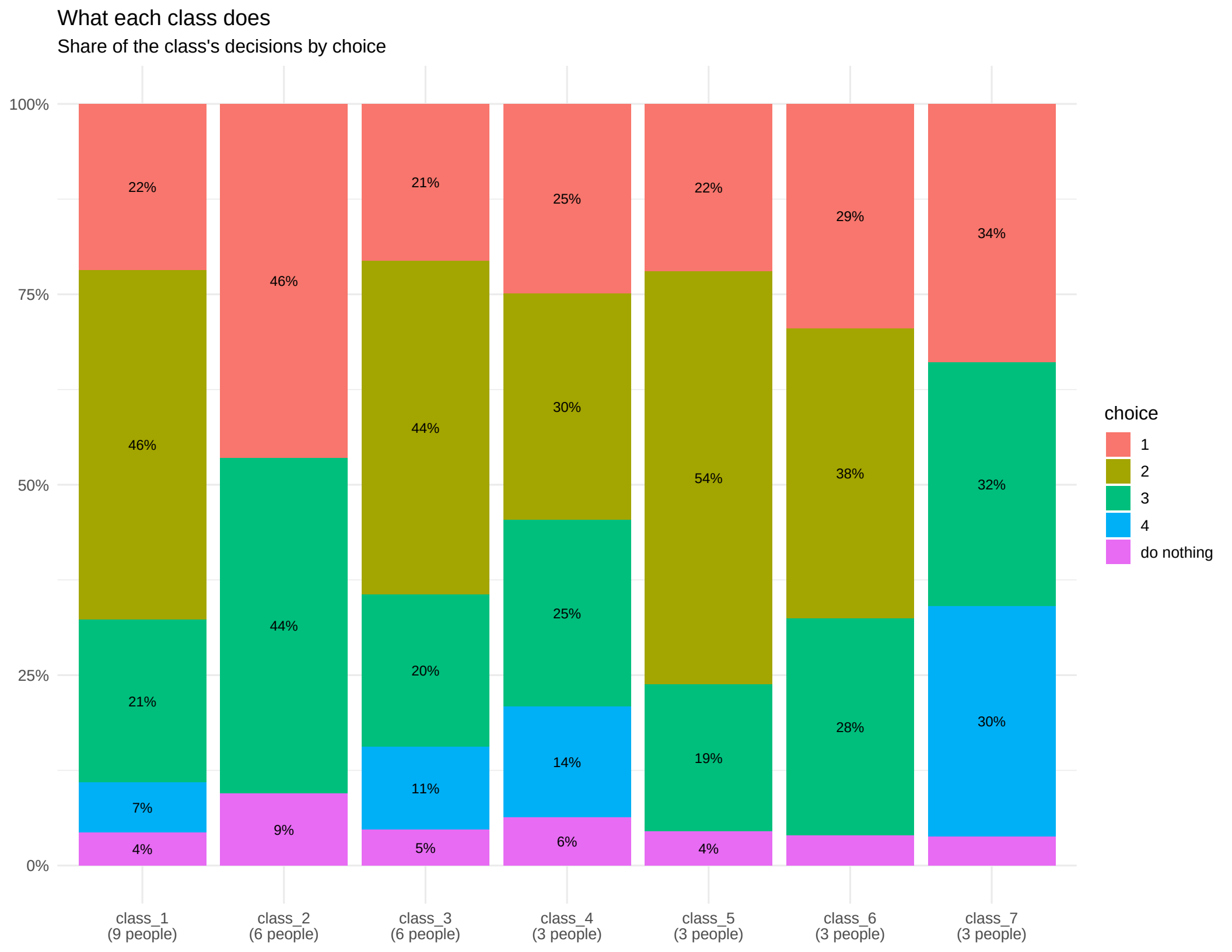
TREATMENTS are session−level settings (e.g. fine probability, cost of protecting). A treatment is USED only if the data vary enough in it, it cannot be confused with another setting, and it improves prediction on whole sessions it was not fitted on. 'Not used' means this data cannot show an effect, not that there is none.

Features named like water?TRUE are yes/no state; names ending in a level (e.g. ownedBy1) are categories compared with the left−out level.

With few sessions or few people per class, treat every picture as a description of this sample, not a finding about farmers in general.

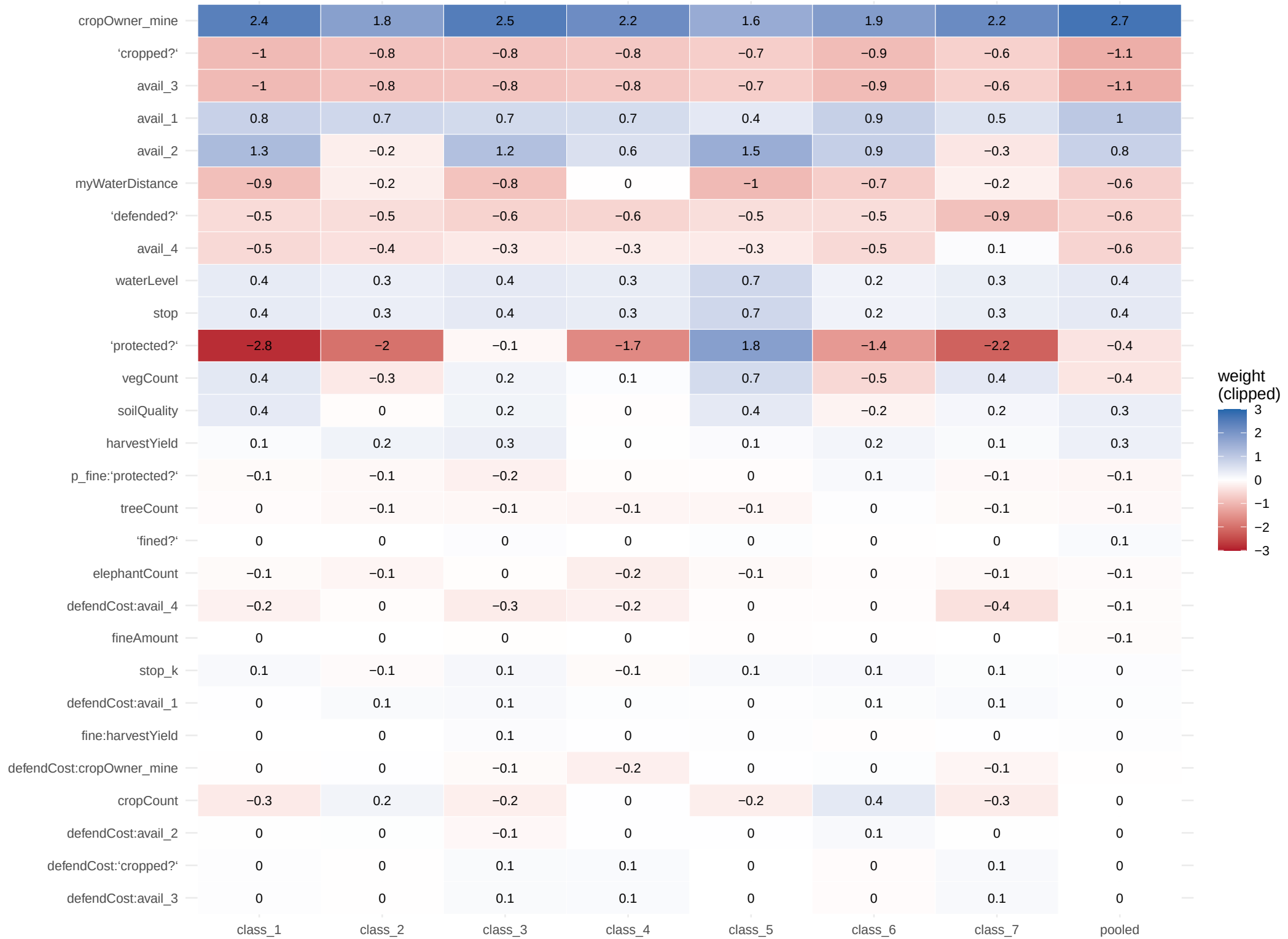

What people chose
Recorded decisions by choice, all sessions
2
1
3
4
do nothing
7676 (36%)
5531 (26%)
5288 (25%)
1722 (8%)
1054 (5%)
0
2500
5000
7500
decisions

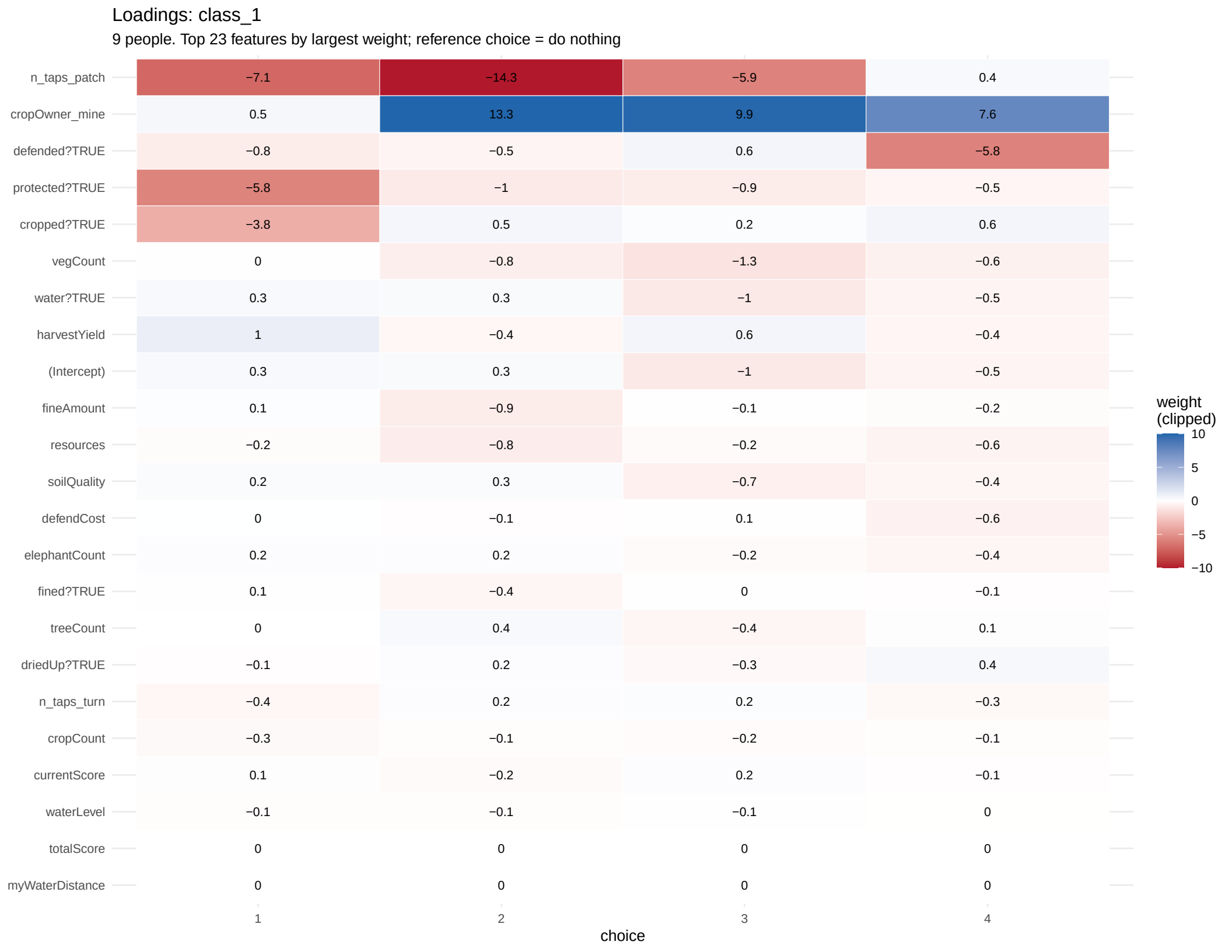

How much data each person contributes
Dashed line = minimum (20) for an individual fit
Chen_114
Dina_289
Kofi_170
Sef_226
Fatu_135
Lena_177
Hana_317
Pia_205
Cato_282
Wen_254
Vik_247
Eli_128
Ana_100
Uma_240
Faris_303
Nia_191
Omar_198
Dara_121
Ivo_324
Hugo_149
Bima_275
Ines_156
Abel_268
Jomo_163
Gus_310
Tavi_233
Ewa_296
Quin_212
Zane_261
Musa_184
Gita_142
Bo_107
Rosa_219
0
500
1000
1500
decisions
class
class_1
class_2
class_3
class_4
class_5
class_6
class_7

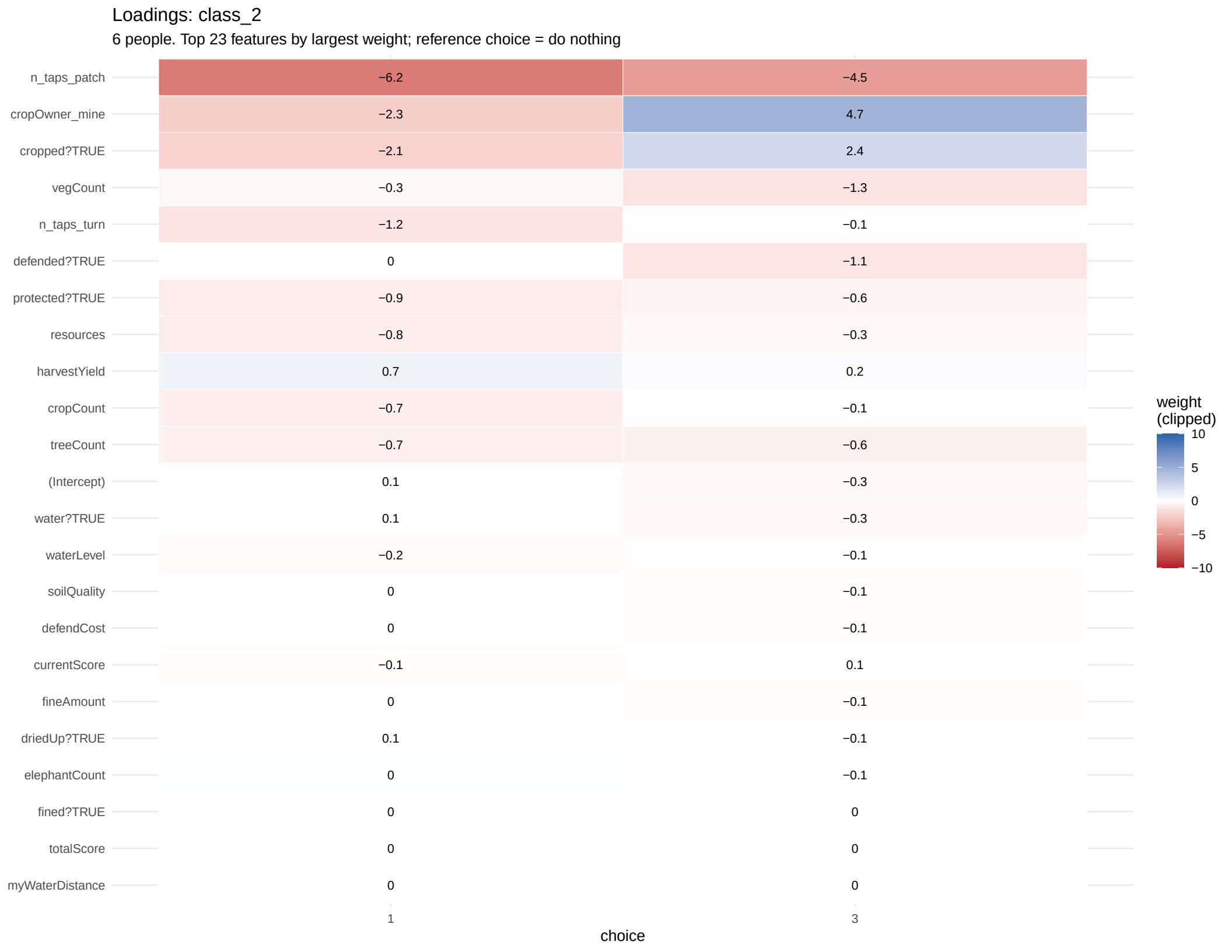

What each class does
Share of the class's decisions by choice
100%
75%
50%
25%
0%
22%
46%
21%
7%
4%
46%
44%
9%
21%
44%
20%
11%
5%
25%
30%
25%
14%
6%
22%
54%
19%
4%
29%
38%
28%
34%
32%
30%
class_1
(9 people)
class_2
(6 people)
class_3
(6 people)
class_4
(3 people)
class_5
(3 people)
class_6
(3 people)
class_7
(3 people)
choice
1
2
3
4
do nothing

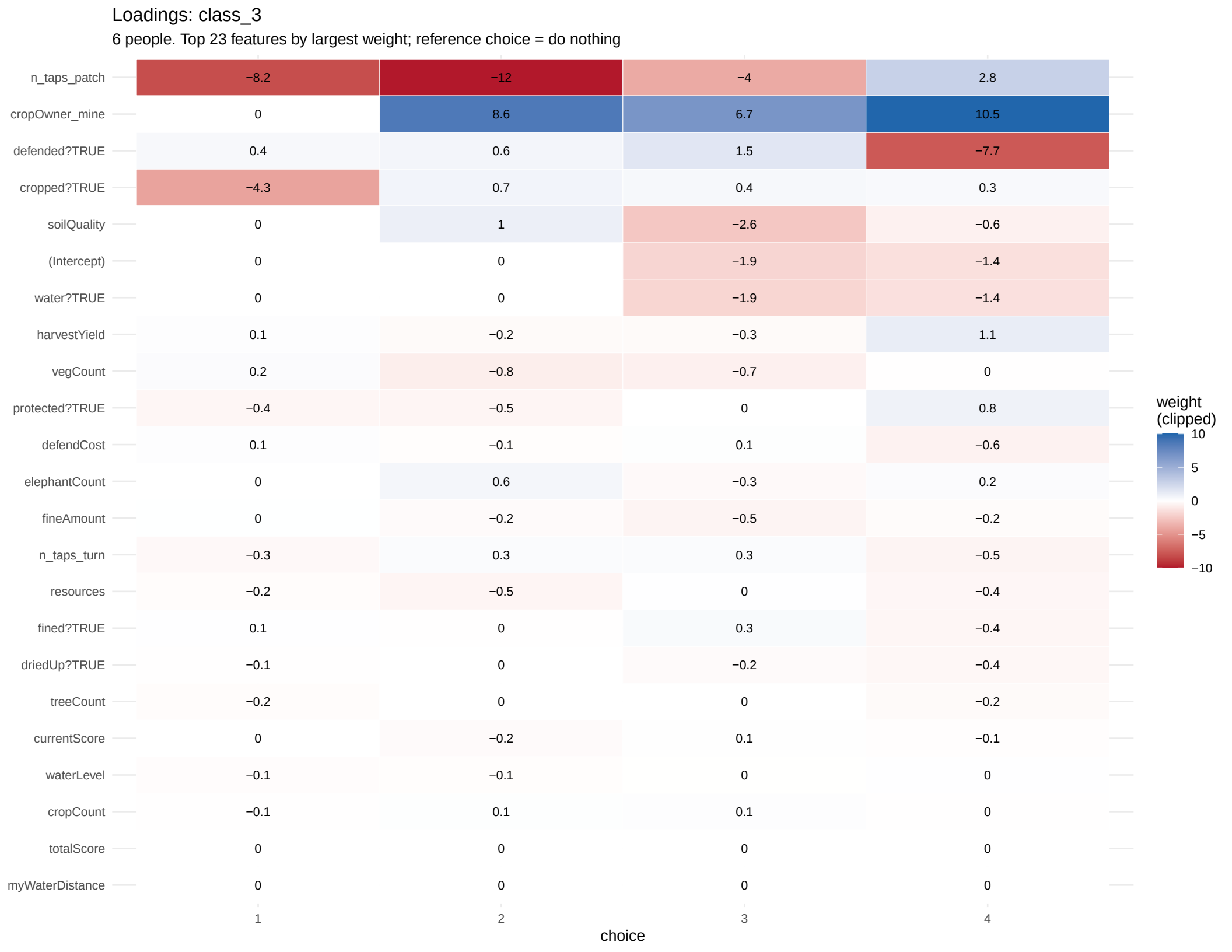
Where each class acts: patch−choice weights

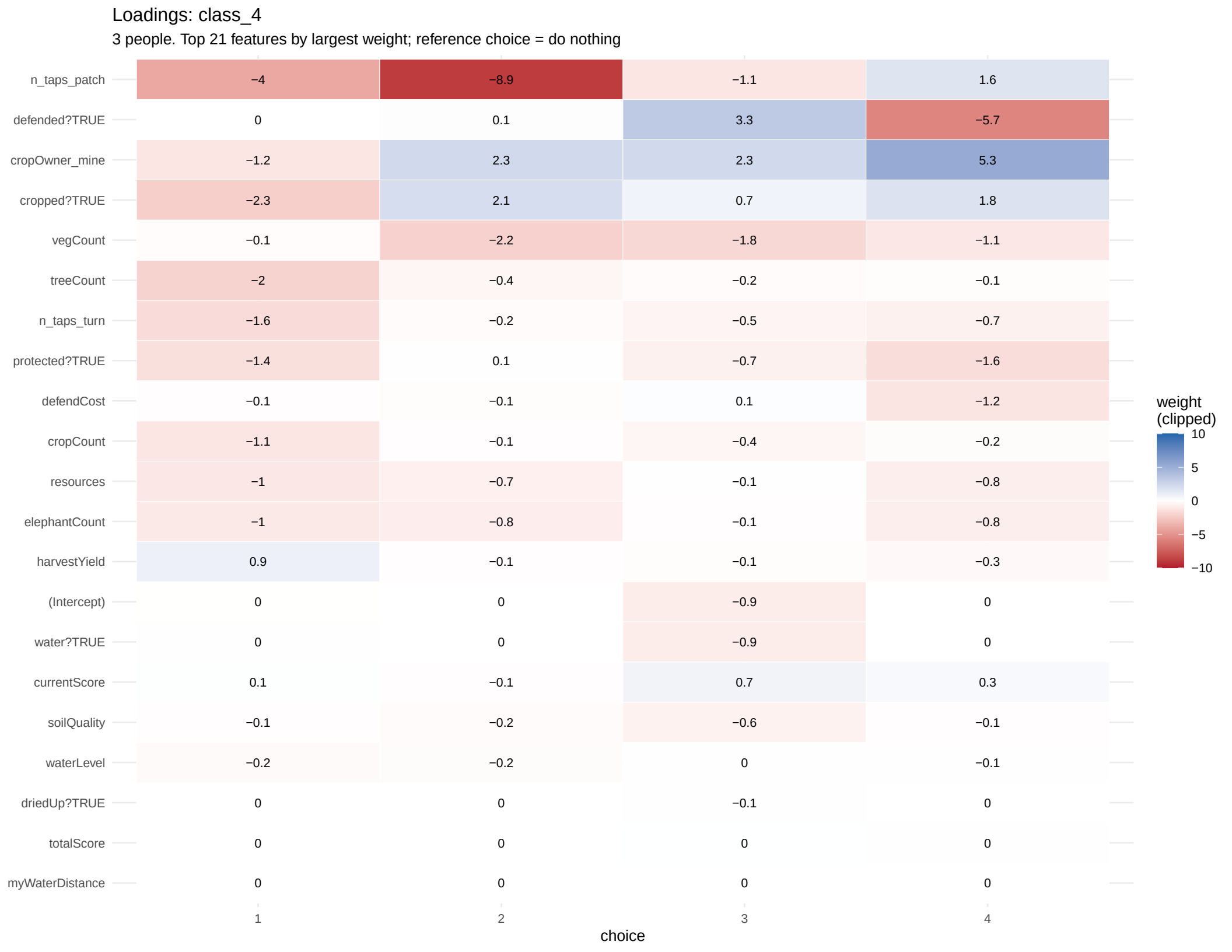
Per standard deviation of each feature. Blue = more likely to be picked next; red = less likely. 'stop' = tendency to end the turn.

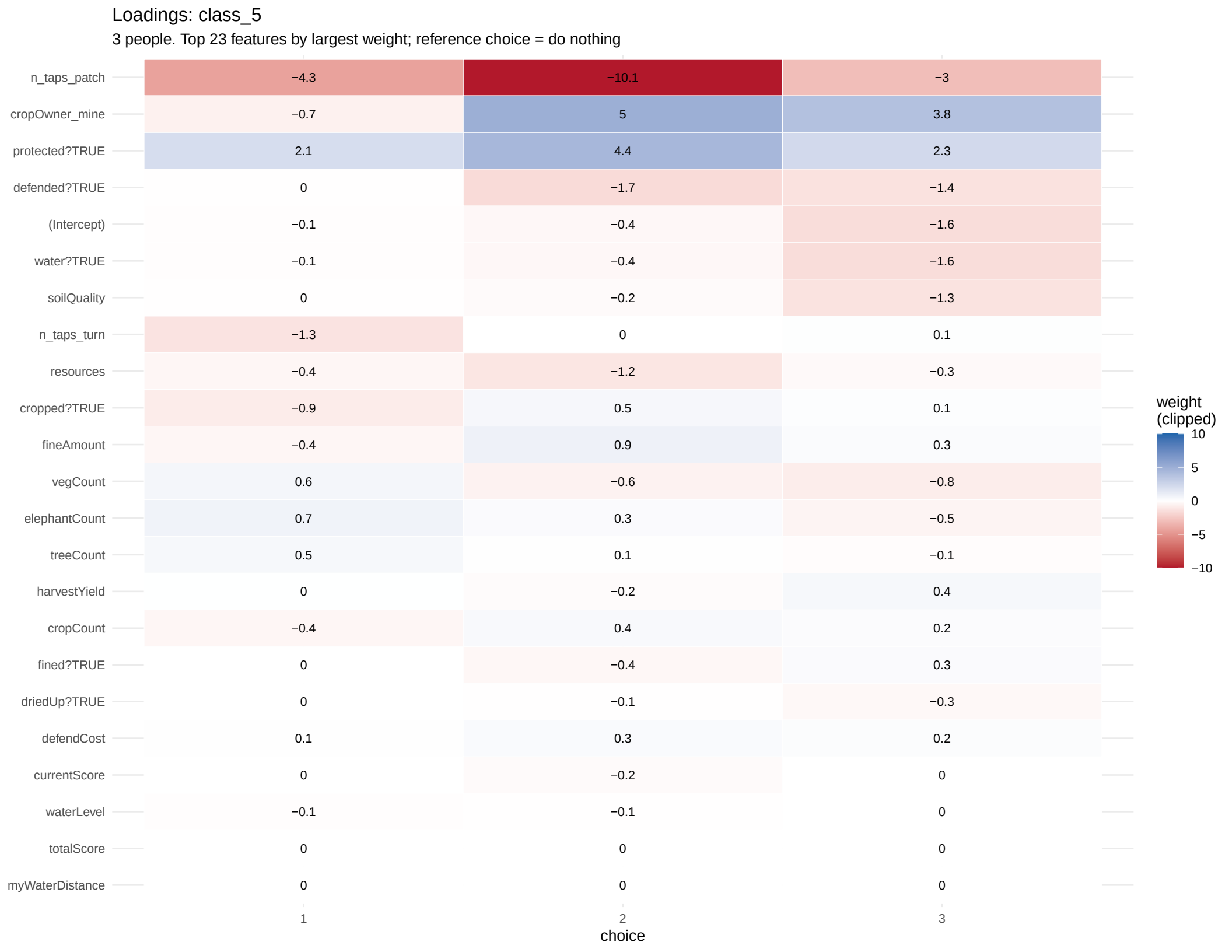


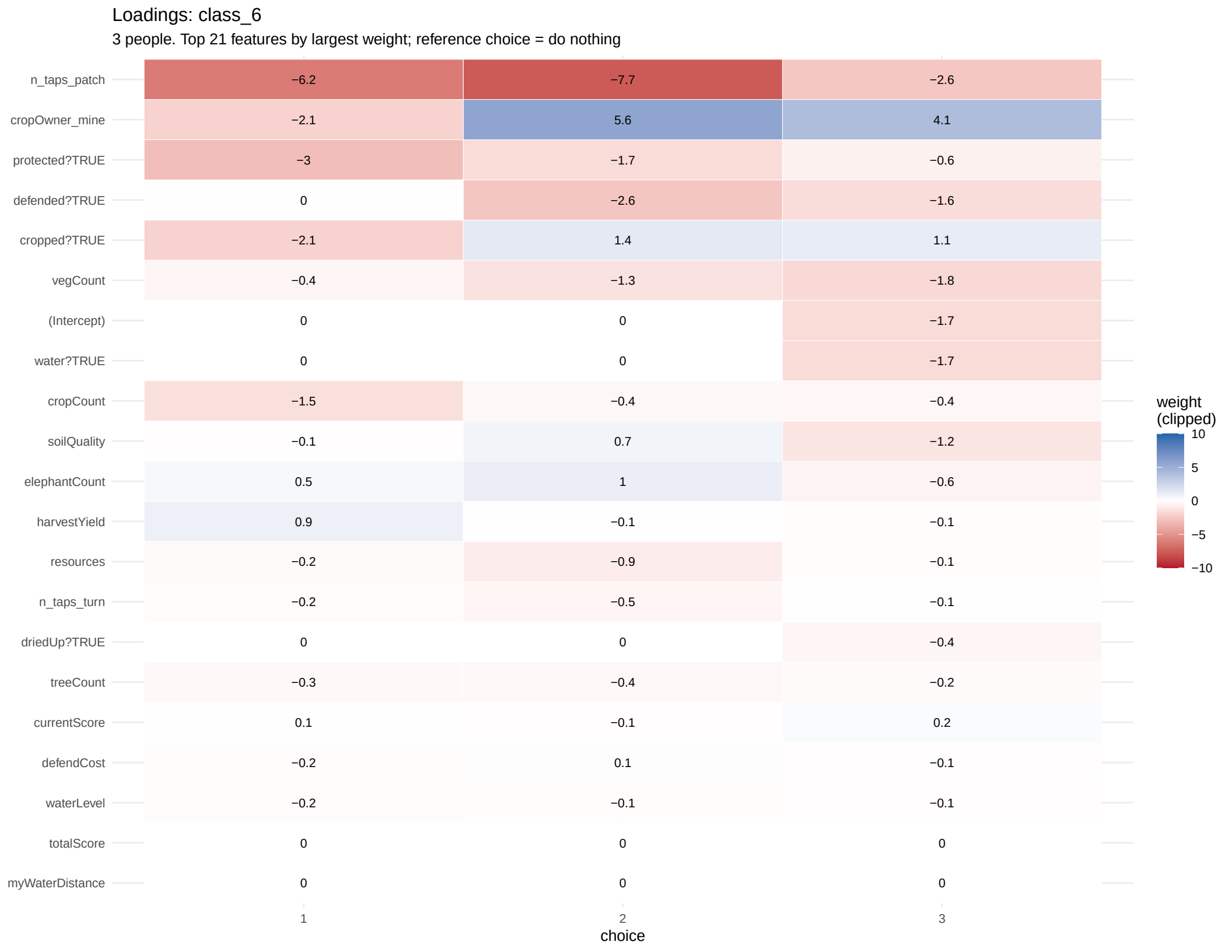

Loadings: class_1
9 people. Top 23 features by largest weight; reference choice = do nothing
n_taps_patch −7.1 −14.3 −5.9 0.4
cropOwner_mine 0.5 13.3 9.9 7.6
defended?TRUE −0.8 −0.5 0.6 −5.8
protected?TRUE −5.8 −1 −0.9 −0.5
cropped?TRUE −3.8 0.5 0.2 0.6
vegCount 0 −0.8 −1.3 −0.6
water?TRUE 0.3 0.3 −1 −0.5
harvestYield 1 −0.4 0.6 −0.4
(Intercept) 0.3 0.3 −1 −0.5
fineAmount 0.1 −0.9 −0.1 −0.2
resources −0.2 −0.8 −0.2 −0.6
soilQuality 0.2 0.3 −0.7 −0.4
defendCost 0 −0.1 0.1 −0.6
elephantCount 0.2 0.2 −0.2 −0.4
fined?TRUE 0.1 −0.4 0 −0.1
treeCount 0 0.4 −0.4 0.1
driedUp?TRUE −0.1 0.2 −0.3 0.4
n_taps_turn −0.4 0.2 0.2 −0.3
cropCount −0.3 −0.1 −0.2 −0.1
currentScore 0.1 −0.2 0.2 −0.1
waterLevel −0.1 −0.1 −0.1 0
totalScore 0 0 0 0
myWaterDistance 0 0 0 0
1
2
3
4
choice
weight (clipped)
10
5
0
−5
−10

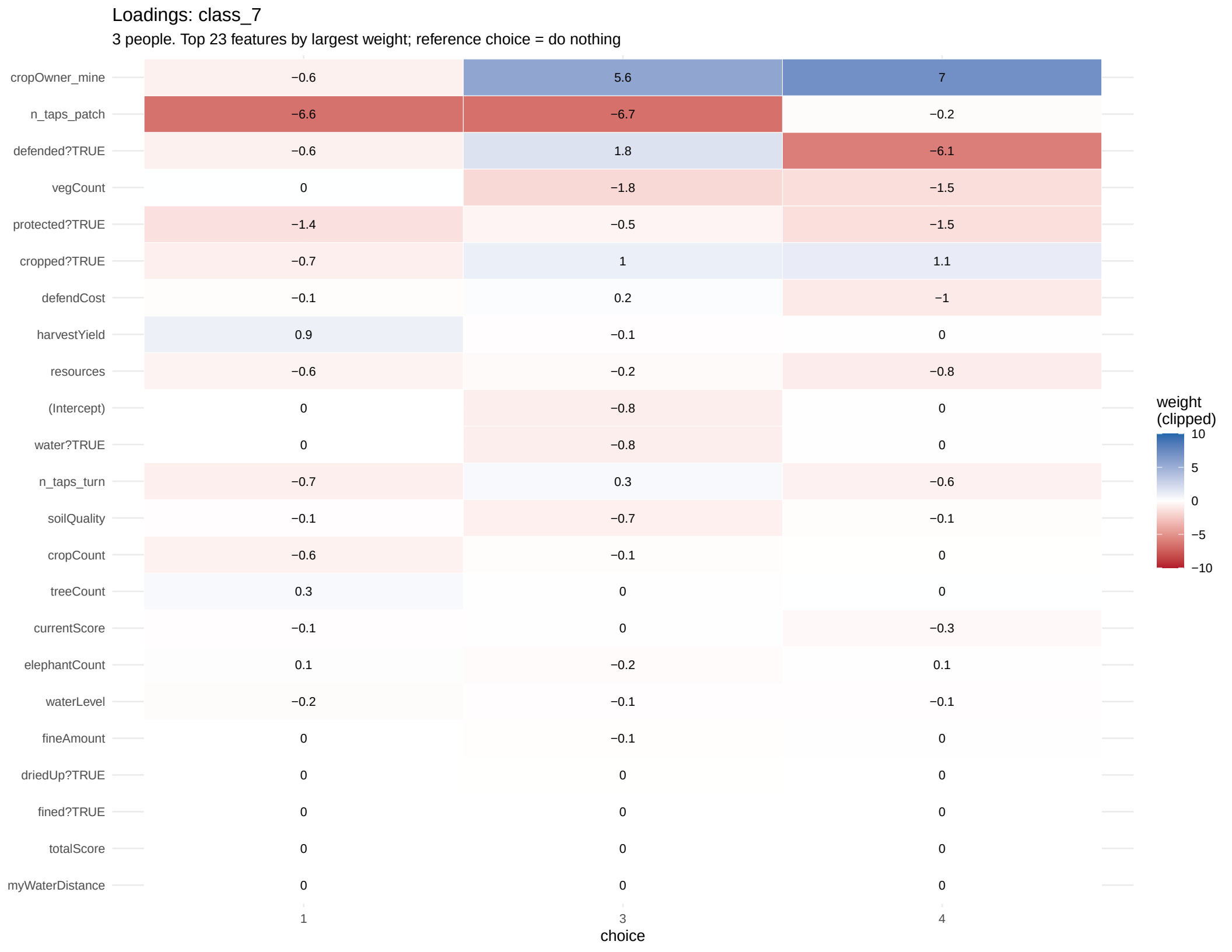

Loadings: class_2
6 people. Top 23 features by largest weight; reference choice = do nothing
n_taps_patch
cropOwner_mine
cropped?TRUE
vegCount
n_taps_turn
defended?TRUE
protected?TRUE
resources
harvestYield
cropCount
treeCount
(Intercept)
water?TRUE
waterLevel
soilQuality
defendCost
currentScore
fineAmount
driedUp?TRUE
elephantCount
fined?TRUE
totalScore
myWaterDistance
−6.2
−4.5
−2.3
4.7
−2.1
2.4
−0.3
−1.3
−1.2
−0.1
0
−1.1
−0.9
−0.6
−0.8
−0.3
0.7
0.2
−0.7
−0.1
−0.7
−0.6
0.1
−0.3
0.1
−0.3
−0.2
−0.1
0
−0.1
0
−0.1
−0.1
0.1
0
−0.1
0.1
−0.1
0
−0.1
0
0
0
0
0
0
1
3
choice
weight (clipped)
10
5
0
−5
−10


Loadings: class_3
6 people. Top 23 features by largest weight; reference choice = do nothing
n_taps_patch −8.2 −12 −4 2.8
cropOwner_mine 0 8.6 6.7 10.5
defended?TRUE 0.4 0.6 1.5 −7.7
cropped?TRUE −4.3 0.7 0.4 0.3
soilQuality 0 1 −2.6 −0.6
(Intercept) 0 0 −1.9 −1.4
water?TRUE 0 0 −1.9 −1.4
harvestYield 0.1 −0.2 −0.3 1.1
vegCount 0.2 −0.8 −0.7 0
protected?TRUE −0.4 −0.5 0 0.8
defendCost 0.1 −0.1 0.1 −0.6
elephantCount 0 0.6 −0.3 0.2
fineAmount 0 −0.2 −0.5 −0.2
n_taps_turn −0.3 0.3 0.3 −0.5
resources −0.2 −0.5 0 −0.4
fined?TRUE 0.1 0 0.3 −0.4
driedUp?TRUE −0.1 0 −0.2 −0.4
treeCount −0.2 0 0 −0.2
currentScore 0 −0.2 0.1 −0.1
waterLevel −0.1 −0.1 0 0
cropCount −0.1 0.1 0.1 0
totalScore 0 0 0 0
myWaterDistance 0 0 0 0
1 2 3 4
choice
weight (clipped)
10
5
0
−5
−10

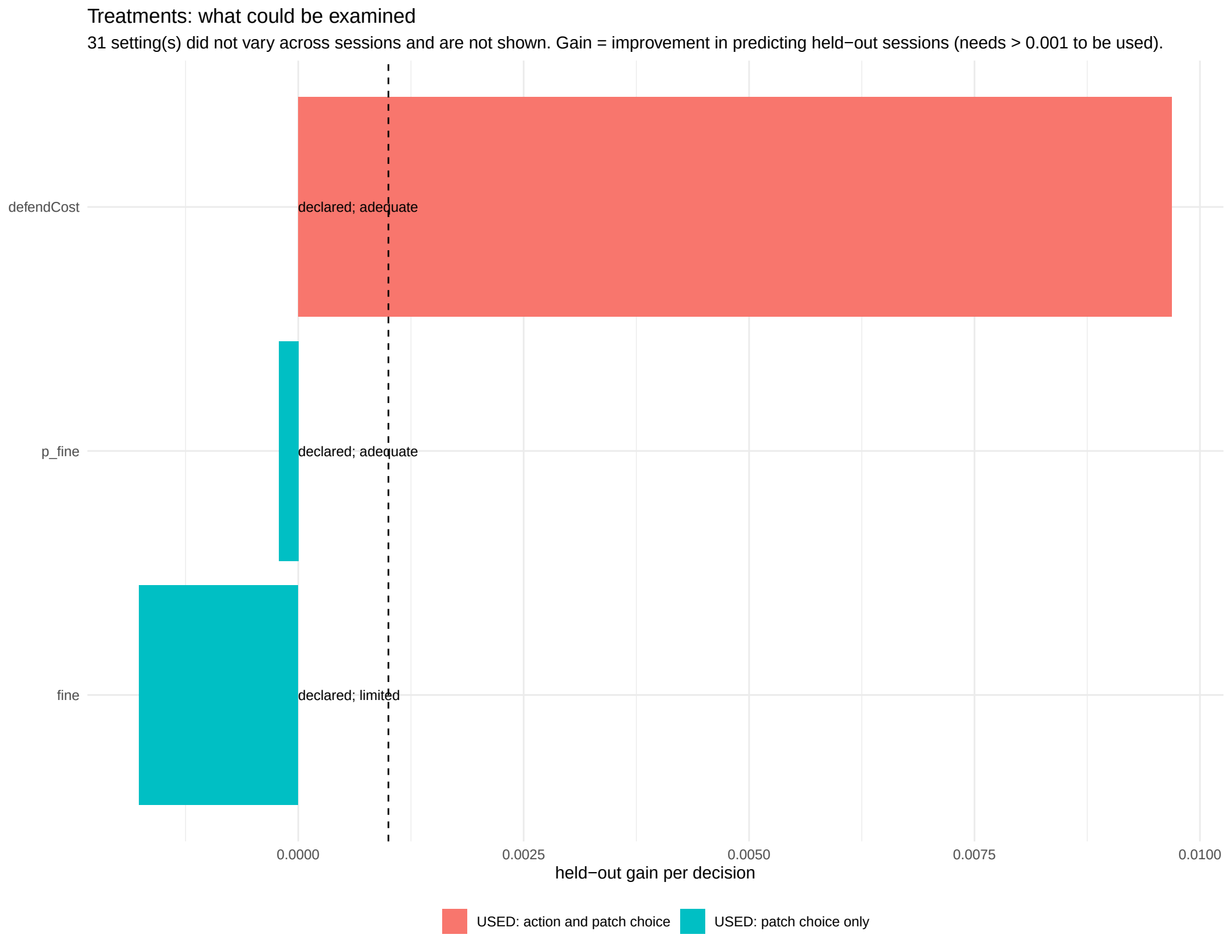

Loadings: class_4
3 people. Top 21 features by largest weight; reference choice = do nothing
n_taps_patch −4 −8.9 −1.1 1.6
defended?TRUE 0 0.1 3.3 −5.7
cropOwner_mine −1.2 2.3 2.3 5.3
cropped?TRUE −2.3 2.1 0.7 1.8
vegCount −0.1 −2.2 −1.8 −1.1
treeCount −2 −0.4 −0.2 −0.1
n_taps_turn −1.6 −0.2 −0.5 −0.7
protected?TRUE −1.4 0.1 −0.7 −1.6
defendCost −0.1 −0.1 0.1 −1.2
cropCount −1.1 −0.1 −0.4 −0.2
resources −1 −0.7 −0.1 −0.8
elephantCount −1 −0.8 −0.1 −0.8
harvestYield 0.9 −0.1 −0.1 −0.3
(Intercept) 0 0 −0.9 0
water?TRUE 0 0 −0.9 0
currentScore 0.1 −0.1 0.7 0.3
soilQuality −0.1 −0.2 −0.6 −0.1
waterLevel −0.2 −0.2 0 −0.1
driedUp?TRUE 0 0 −0.1 0
totalScore 0 0 0 0
myWaterDistance 0 0 0 0
1 2 3 4
choice
weight (clipped)
10
5
0
−5
−10

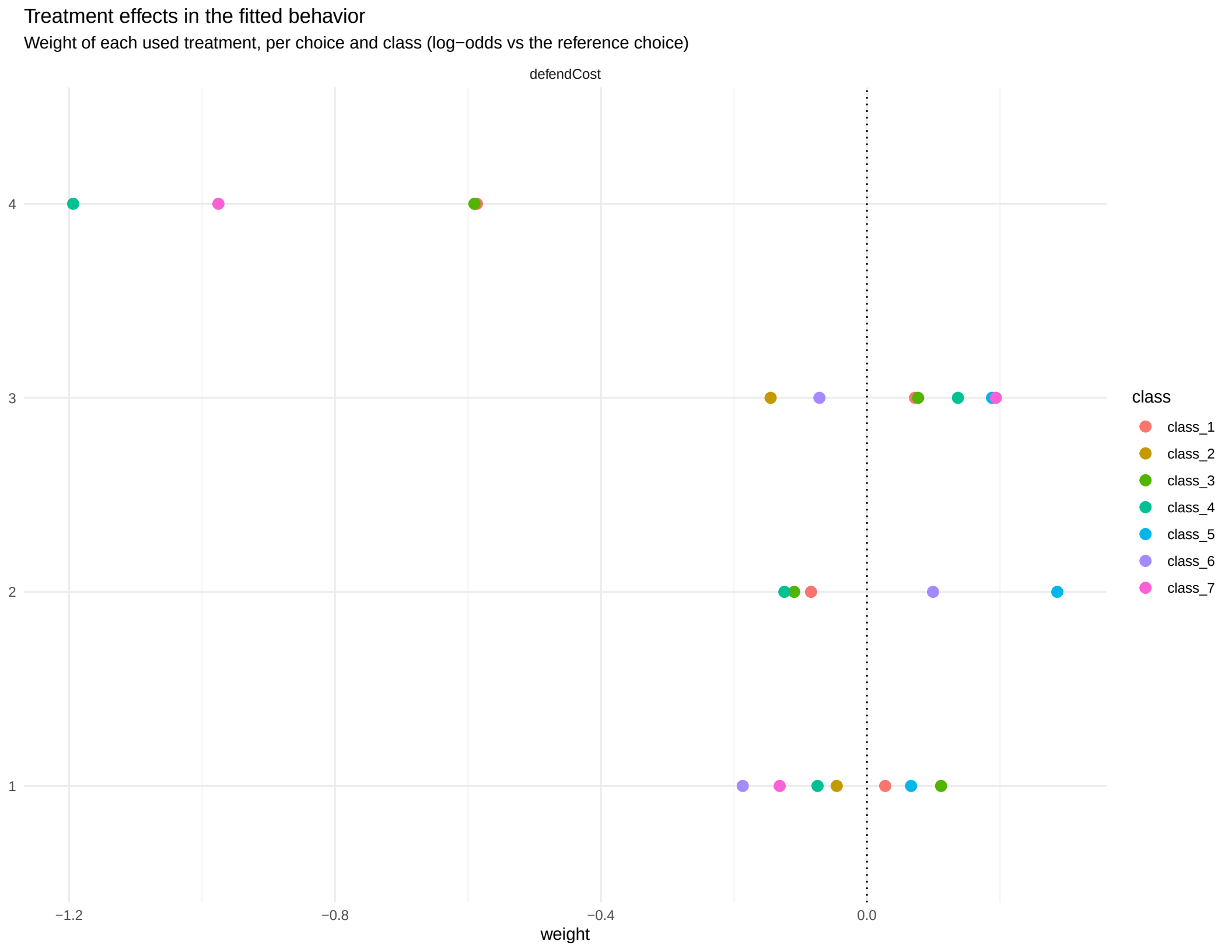

Loadings: class_5
3 people. Top 23 features by largest weight; reference choice = do nothing
n_taps_patch −4.3 −10.1 −3
cropOwner_mine −0.7 5 3.8
protected?TRUE 2.1 4.4 2.3
defended?TRUE 0 −1.7 −1.4
(Intercept) −0.1 −0.4 −1.6
water?TRUE −0.1 −0.4 −1.6
soilQuality 0 −0.2 −1.3
n_taps_turn −1.3 0 0.1
resources −0.4 −1.2 −0.3
cropped?TRUE −0.9 0.5 0.1
fineAmount −0.4 0.9 0.3
vegCount 0.6 −0.6 −0.8
elephantCount 0.7 0.3 −0.5
treeCount 0.5 0.1 −0.1
harvestYield 0 −0.2 0.4
cropCount −0.4 0.4 0.2
fined?TRUE 0 −0.4 0.3
driedUp?TRUE 0 −0.1 −0.3
defendCost 0.1 0.3 0.2
currentScore 0 −0.2 0
waterLevel −0.1 −0.1 0
totalScore 0 0 0
myWaterDistance 0 0 0
1
2
3
choice
weight (clipped)
10
5
0
−5
−10


Loadings: class_6
3 people. Top 21 features by largest weight; reference choice = do nothing
n_taps_patch −6.2 −7.7 −2.6
cropOwner_mine −2.1 5.6 4.1
protected?TRUE −3 −1.7 −0.6
defended?TRUE 0 −2.6 −1.6
cropped?TRUE −2.1 1.4 1.1
vegCount −0.4 −1.3 −1.8
(Intercept) 0 0 −1.7
water?TRUE 0 0 −1.7
cropCount −1.5 −0.4 −0.4
soilQuality −0.1 0.7 −1.2
elephantCount 0.5 1 −0.6
harvestYield 0.9 −0.1 −0.1
resources −0.2 −0.9 −0.1
n_taps_turn −0.2 −0.5 −0.1
driedUp?TRUE 0 0 −0.4
treeCount −0.3 −0.4 −0.2
currentScore 0.1 −0.1 0.2
defendCost −0.2 0.1 −0.1
waterLevel −0.2 −0.1 −0.1
totalScore 0 0 0
myWaterDistance 0 0 0
1
2
3
choice
weight (clipped)
10
5
0
−5
−10


Loadings: class_7
3 people. Top 23 features by largest weight; reference choice = do nothing
cropOwner_mine −0.6 5.6 7
n_taps_patch −6.6 −6.7 −0.2
defended?TRUE −0.6 1.8 −6.1
vegCount 0 −1.8 −1.5
protected?TRUE −1.4 −0.5 −1.5
cropped?TRUE −0.7 1 1.1
defendCost −0.1 0.2 −1
harvestYield 0.9 −0.1 0
resources −0.6 −0.2 −0.8
(Intercept) 0 −0.8 0
water?TRUE 0 −0.8 0
n_taps_turn −0.7 0.3 −0.6
soilQuality −0.1 −0.7 −0.1
cropCount −0.6 −0.1 0
treeCount 0.3 0 0
currentScore −0.1 0 −0.3
elephantCount 0.1 −0.2 0.1
waterLevel −0.2 −0.1 −0.1
fineAmount 0 −0.1 0
driedUp?TRUE 0 0 0
fined?TRUE 0 0 0
totalScore 0 0 0
myWaterDistance 0 0 0
1
3
4
choice
weight (clipped)
10
5
0
−5
−10


What separates the classes
The 20 (choice | feature) pairs whose weights differ most between classes
2 | cropOwner_mine
1 | protected?TRUE
3 | cropOwner_mine
2 | n_taps_patch
2 | protected?TRUE
3 | n_taps_patch
4 | cropOwner_mine
3 | defended?TRUE
1 | n_taps_patch
1 | cropped?TRUE
2 | defended?TRUE
3 | protected?TRUE
4 | n_taps_patch
1 | cropOwner_mine
1 | treeCount
3 | soilQuality
4 | protected?TRUE
3 | cropped?TRUE
4 | defended?TRUE
2 | elephantCount
−15
−10
−5
0
5
10
weight
class
class_1
class_2
class_3
class_4
class_5
class_6
class_7


Treatments: what could be examined
31 setting(s) did not vary across sessions and are not shown. Gain = improvement in predicting held−out sessions (needs > 0.001 to be used).
defendCost
declared; adequate
p_fine
declared; adequate
fine
declared; limited
0.0000
0.0025
0.0050
0.0075
0.0100
held−out gain per decision
USED: action and patch choice
USED: patch choice only


Treatment effects in the fitted behavior
Weight of each used treatment, per choice and class (log−odds vs the reference choice)
defendCost
4
3
2
1
−1.2
−0.8
−0.4
0.0
weight
class
class_1
class_2
class_3
class_4
class_5
class_6
class_7

Who is in which class (1 of 1)

```
person sessions decisions class

Abel_268 8 482 class_1

Ana_100 9 653 class_1

Chen_114 15 1465 class_1

Faris_303 9 602 class_1

Hugo_149 10 552 class_1

Lena_177 12 895 class_1

Nia_191 7 587 class_1

Sef_226 12 1033 class_1

Wen_254 12 800 class_1

Bo_107 7 238 class_2

Gus_310 12 458 class_2

Ines_156 13 507 class_2

Musa_184 6 311 class_2

Tavi_233 9 412 class_2

Zane_261 8 314 class_2

Cato_282 10 800 class_3

Dina_289 13 1254 class_3

Eli_128 14 770 class_3

Fatu_135 12 1002 class_3

Pia_205 11 855 class_3
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